AI Playbook and Toolkit for Public Health Departments

Public Health Data Modernization and AI Architecture

ARC 200

Artificial intelligence does not operate separately from the public health data environment. Every AI-supported workflow depends on the quality, structure, timeliness, governance, and security of the systems that supply data to it. For public health agencies, this means that AI readiness is inseparable from data modernization. A department cannot responsibly deploy AI for surveillance, case investigation, forecasting, summarization, resource allocation, public communication, or operational decision support unless it understands the systems, data flows, authorities, and human decisions that shape the workflow. This module introduces public health data modernization as the technical foundation for AI. Learners examine how data move from source systems into public health systems, how those data are transformed and used, and where AI might be introduced safely. The goal is not to teach every learner to design enterprise architecture. The goal is to help informaticians, epidemiologists, technologists, program leaders, and governance participants understand enough about architecture to ask the right questions before AI tools are selected, procured, configured, or deployed. By the end of the module, learners should understand that an AI tool is only one component of a broader public health system. The surrounding architecture includes data sources, interfaces, terminology standards, identity and access controls, cloud or on-premises infrastructure, analytics environments, workflow applications, audit logs, governance processes, monitoring, and human review. Responsible AI planning begins by making that system visible.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Explain why AI readiness depends on public health data modernization, not only on access to an AI tool.
- Describe the major components of an AI-enabled public health architecture, including source systems, data exchange, data transformation, AI services, workflow applications, access controls, audit logs, and monitoring.
- Identify the systems of record, source systems, data flows, and human decision points in a public health workflow.
- Assess where AI could support a workflow and where human review, legal authority, epidemiologic judgment, or community context must remain central.
- Identify architectural risks that can affect AI safety, including incomplete data, unclear authority, excessive system access, weak monitoring, and lack of documentation.
- Produce an AI-enabled workflow architecture map and readiness memo that can support governance review, implementation planning, procurement, or vendor evaluation.

Module Overview

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By the end of the module, learners should understand that an AI tool is only one component of a broader public health system. The surrounding architecture includes data sources, interfaces, terminology standards, identity and access controls, cloud or on-premises infrastructure, analytics environments, workflow applications, audit logs, governance processes, monitoring, and human review. Responsible AI planning begins by making that system visible.

Why This Topic Matters for Public Health AI

Public health agencies often feel pressure to adopt AI tools quickly, especially when staff are overwhelmed, reporting demands are increasing, and data modernization funding is available. AI tools may appear to offer immediate relief by summarizing records, drafting communications, classifying incoming reports, identifying possible outbreaks, or automating repetitive tasks. However, AI can create new risks when it is layered on top of fragmented or poorly understood systems. A model that summarizes case reports may produce incomplete summaries if the source data are missing key fields. A triage algorithm may reinforce inequities if the underlying system receives more complete data from some providers or communities than others. An agentic workflow may trigger inappropriate actions if the system does not clearly define what the AI tool is allowed to read, write, update, or escalate.

Data modernization provides the foundation for reducing these risks. Modernization is not just a technology upgrade. It includes better data exchange, stronger governance, improved data quality, workforce capacity, reusable services, interoperability, security, and more timely access to information. AI readiness should therefore be treated as a modernization outcome. If an agency cannot describe where the data come from, how they are transformed, who is authorized to use them, and how decisions are documented, then the agency is not yet ready to use AI for that workflow.

This matters especially in public health because technical errors can become operational errors. A poorly designed AI workflow could delay case investigation, misclassify a reportable condition, summarize a record inaccurately, miss a vulnerable population, or create a public communication that overstates evidence. Architecture review helps prevent these errors by making the full system visible before AI is introduced.

Core Concepts

Public health data modernization. Public health data modernization is the process of improving the systems, standards, infrastructure, governance, and workforce capacity needed to collect, exchange, manage, analyze, and use public health data. Modernization includes technical components such as APIs, cloud platforms, data warehouses, terminology services, analytics environments, and interoperability standards. It also includes operational components such as data stewardship, governance review, privacy controls, workforce training, procurement discipline, and performance monitoring. AI should be understood as one possible modernization capability, not as a replacement for the modernization work itself.

AI architecture. AI architecture refers to the set of systems and processes that allow an AI tool to function within a public health workflow. This includes the source data, data pipelines, model or AI service, user interface, security controls, audit logs, monitoring process, and human review steps. In a mature architecture, each component has a defined role. Staff know which system is the source of truth, which data are allowed to be used, how outputs are reviewed, and what happens when the AI tool produces an uncertain, incomplete, or incorrect result.

Source system. A source system is the system where data originate or are first captured. In public health, source systems may include electronic health records, laboratories, immunization information systems, vital records systems, syndromic surveillance feeds, environmental health systems, claims datasets, community resource directories, or manually entered program data. Source systems differ in quality, timeliness, completeness, permissions, and terminology. These differences directly affect whether downstream AI outputs can be trusted.

System of record. A system of record is the official system or process where public health information, decisions, and actions are maintained. AI outputs should not automatically become the official record unless the agency has explicitly approved that use. For example, an AI tool may draft a case summary, but the final case note may still need to be reviewed, edited, and saved by authorized staff in the surveillance system. The distinction between AI-generated support and official public health documentation is essential for accountability, public records management, legal review, and incident response.

Data lineage and audit logs. Data lineage documents where data came from, how they were changed, and where they were used. Audit logs record actions taken by users, systems, or automated tools. Together, lineage and audit logs help agencies explain why an AI output was generated, which data were used, who reviewed it, whether it was accepted or edited, and whether downstream action was taken. Without these records, agencies may struggle to investigate errors or demonstrate responsible oversight.

Public Health Example

Consider a health department that wants to use AI to summarize incoming electronic case reports for disease investigators. The AI tool may be able to produce a concise summary of symptoms, laboratory results, diagnosis, treatment, provider information, exposure information, and missing fields. However, the safety of that workflow depends on the surrounding architecture. The eCR data must be received correctly, routed to the right jurisdiction, parsed into the surveillance system, associated with the correct person or case, and made available only to authorized users.

The AI tool also needs clear boundaries. It should know which fields it can use and whether it is allowed to access narrative text, laboratory results, demographic data, prior case history, or external reference material. If the tool uses retrieval-augmented generation, the agency must know which policies, disease protocols, or guidance documents are included in the retrieval source. If the tool can create tasks or update case records, the agency must define whether those actions require human approval.

The system also needs a clear human review point. The investigator should be able to see the source information behind the AI summary, identify unsupported statements, correct errors, and document the final interpretation in the surveillance system. The agency should log when the AI summary was generated, which data were used, who reviewed it, and whether the output was accepted, edited, rejected, or escalated. Without that architecture, the AI summary could appear efficient while creating hidden risks.

Technical Deep Dive

A public health AI architecture usually begins with source systems. These may include electronic health records, laboratories, immunization information systems, vital records systems, syndromic surveillance feeds, case management systems, health information exchanges, claims datasets, community resource directories, environmental health systems, or manually entered program data. Each source system has its own data quality profile, update frequency, terminology conventions, permissions, and limitations. Before AI can be used safely, the agency needs to understand these characteristics.

The next layer is data exchange and ingestion. Data may arrive through HL7 v2 messages, CDA documents, FHIR APIs, flat files, web portals, batch uploads, or manual entry. Each method creates different risks. A real-time interface may fail silently if monitoring is weak. A batch file may be delayed. A manual upload may introduce formatting errors. A FHIR API may expose only certain data elements. AI planning must account for how data enter the workflow and how the agency knows whether the data are complete, current, and authorized for the intended use.

The third layer is data transformation and storage. Public health data are often cleaned, mapped, deduplicated, linked, normalized, geocoded, or summarized before they are used. These steps are essential, but they can also introduce errors. For example, local laboratory codes may be mapped incorrectly to standard codes. Duplicate person records may be merged incorrectly. Addresses may be geocoded to the wrong jurisdiction. Race, ethnicity, language, disability, or housing status fields may be missing or inconsistently captured. If an AI tool uses transformed data, the agency must be able to trace how those data were created and whether transformation decisions affect the output.

The fourth layer is the AI service or model. This may be a predictive model, natural language processing tool, generative AI assistant, retrieval-augmented generation system, rules-based automation, or agentic workflow. The model should have a defined intended use and explicit limits. A tool designed to summarize case notes should not silently make eligibility decisions. A tool designed to classify complaints should not automatically trigger enforcement action unless that use has been reviewed and approved. A tool designed to support resource allocation should not be used without subgroup performance testing and equity review.

The final layers are workflow integration, monitoring, and governance. AI outputs must appear where staff can use them appropriately and should preserve the ability to inspect source information, correct errors, override outputs, document decisions, and escalate concerns. Monitoring should include technical measures such as uptime, latency, feed failures, missing data, error rates, and drift, as well as operational measures such as user overrides, complaints, incident reports, equity concerns, timeliness, staff burden, and downstream public health outcomes.

Risks, Failure Modes, and Guardrails

Architectural invisibility. If staff cannot explain how data move through the system, they may not recognize when an AI output is based on incomplete, delayed, duplicated, or incorrectly transformed data. A practical guardrail is to require a workflow and data flow map before AI use is approved.

Unclear authority. AI tools may produce recommendations, summaries, or classifications that look official, even when they have not been reviewed. A practical guardrail is to define which system remains the official system of record and which AI outputs require human approval before they can be used for public health action.

Excessive access. AI tools connected through APIs, plugins, or workflow integrations may be able to retrieve, summarize, transmit, or modify sensitive data. A practical guardrail is to apply least-privilege access, role-based permissions, audit logging, sandbox testing, and explicit approval for any write-back function.

Poor monitoring and equity harm. AI tools can degrade over time when data patterns, reporting sources, terminology, policies, or workflows change. If the architecture relies on data sources that underrepresent some communities, the tool may also produce inequitable outputs. Practical guardrails include pre-deployment subgroup review, monitoring indicators, user override tracking, and clear authority to pause or modify use.

Application to AI-Supported Workflows

For generative AI, architecture determines what source material the tool can access, whether it can cite sources, whether it can use sensitive data, and how outputs are reviewed. A generative AI tool used to summarize policy documents has different architectural requirements than one used to summarize case notes or draft public health advisories. The more sensitive or operational the use case, the stronger the controls should be.

For predictive analytics, architecture determines whether the model receives timely and consistent data, whether the prediction is displayed in context, and whether users understand the limits of the score. A risk score should not become an automatic decision unless the agency has explicitly approved that use, validated the model, reviewed subgroup performance, and established human review procedures.

For retrieval-augmented generation and agentic AI, architecture determines which documents or systems are available, how permissions are enforced, and what actions the AI can take. An agent that can create tasks, send messages, update records, or trigger alerts must operate within strict boundaries. The agency must define allowed actions, prohibited actions, approval points, logs, rollback procedures, and incident response triggers before deployment.

Reflection Questions

Which public health workflow in your agency would be most helpful to map before considering AI?

What are the source systems, systems of record, and downstream outputs for that workflow?

Where are data transformed, linked, summarized, interpreted, or manually corrected?

Which parts of the workflow require legal, epidemiologic, clinical, communications, equity, privacy, or leadership review?

What could go wrong if AI were added without changing the architecture?

Which monitoring indicators would show whether the AI-supported workflow is working safely and equitably?

Practical Exercise

Create an AI-enabled workflow architecture map for one public health workflow. Choose a workflow that is real or realistic for your agency, such as case report intake, lab result triage, immunization record review, community complaint classification, outbreak summary drafting, public inquiry response, referral management, syndromic surveillance review, or resource allocation.

Your map should include the source systems, data exchange methods, transformation steps, system of record, AI-supported task, user interface, human review point, output or decision, audit log, monitoring process, and escalation pathway. After completing the map, identify at least five readiness gaps that should be addressed before implementation.

Expected Artifact or Evidence

The expected artifact is an AI-enabled workflow architecture map. The evidence should also include a short readiness memo that identifies technical dependencies, governance questions, human review requirements, access control needs, monitoring indicators, and unresolved risks.

A strong completion artifact should be specific enough to support governance review, implementation planning, procurement, or vendor evaluation. It should not simply state that AI may be useful. It should show how the AI-supported workflow would function, what safeguards would be needed, and what questions remain unresolved.

Expected Assignment

- The expected artifact is an AI-enabled workflow architecture map. The evidence should also include a short readiness memo that identifies technical dependencies, governance questions, human review requirements, access control needs, monitoring indicators, and unresolved risks.
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Knowledge Check

- 1. Why should AI planning begin with architecture and workflow mapping?
- 2. What is the role of a system of record in an AI-supported workflow?
- 3. Which risk is most likely when staff cannot explain how data move through a workflow?
- 4. What is an appropriate guardrail for an AI tool that can update a public health system?
- 5. What is strong completion evidence for this module?

References and Resources

- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- CDC Data Modernization Initiative materials: https://www.cdc.gov/data-modernization/php/index.html
- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- ONC interoperability and health IT resources
- Agency data governance, privacy, cybersecurity, and enterprise architecture documentation